Course Name:	**Artificial Intelligence Laboratory**	**Semester:**	**VI**
Date of Performance:	**22 / 01 / 2026**	**DIV/ Batch No:**	**A1**
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Experiment No: 1

Title: Implementation of PROLOG programs

Aim: To implement condition-action rules based agent and Goal based agent using PROLOG program

Objectives of the Experiment:

By the end of this experiment, students will be able to:

1. Understand condition-action rule based agents
2. Implement simple AI agents using PROLOG facts and rules
3. Design a goal-based agent using logical reasoning
4. Compare reflex and goal-based agent behavior

COs to be achieved:

CO1: Design AI solution with appropriate choice of agent architecture

Books/ Journals/ Websites referred:

1. Russell & Norvig, Artificial Intelligence: A Modern Approach
2. PROLOG Programming for Artificial Intelligence – Ivan Bratko
3. SWI-Prolog Documentation

Theory:

Agent programs for simple applications need not be very complicated. They can be based on condition-action rules and still they give better results, though not always rational. The family tree program makes use of a similar concept.

A simple agent program can be defined mathematically as an agent function which maps every possible percepts sequence to a possible action the agent can perform or to a coefficient, feedback element, function or constant that affects eventual actions:

$$F: P* -> A$$

Algorithm for 'Condition-Action Rule Table' Agent function:

functionSIMPLE-REFLEX-AGENT (percept) **returns** an action
Static: *rules,* a set of condition-action rules
State:- INTERPRET-INPUT (percept)
Rule:- *RULE-MATCH (state, rules)*
Action:- *RULE-ACTION [rule]*
Returnaction

This approach follows a table for lookup of condition-action pairs defining all possible condition-action rules necessary to interact in an environment.

Goal Based Agent:

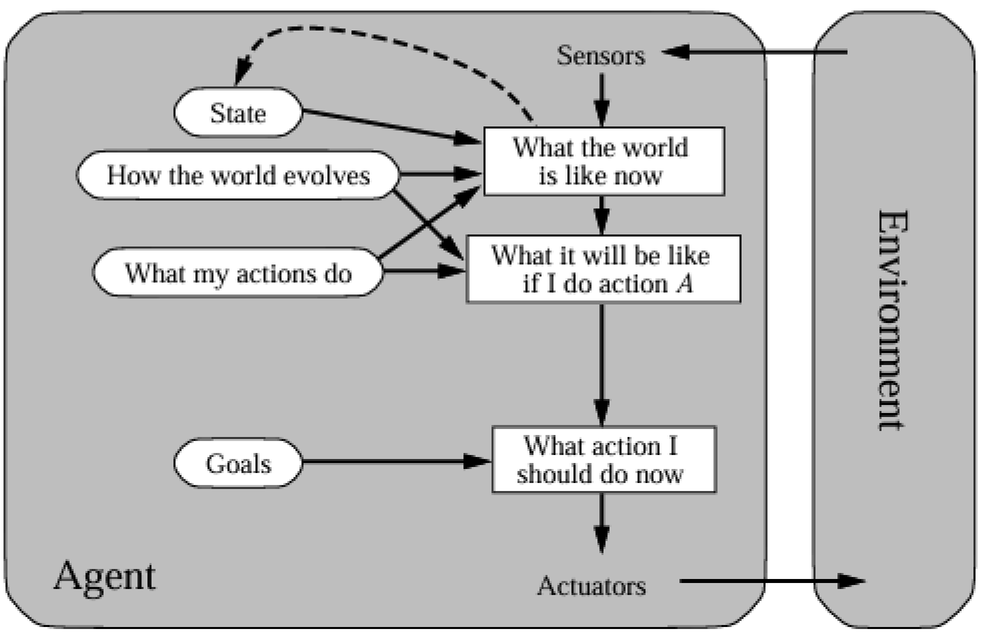


Figure: Goal based agent architecture

The decision making in goal based agents is fundamentally different from the condition-action rules based agents. A goal-based agent, in principle, could reason that if the car in front has its brake lights on, it will slow down. Given the way the world usually evolves, the only action that will achieve the goal of not hitting other cars is to brake. Although the goal-based agent appears less efficient, it is more flexible because the knowledge that supports its decisions is represented explicitly and can be modified.

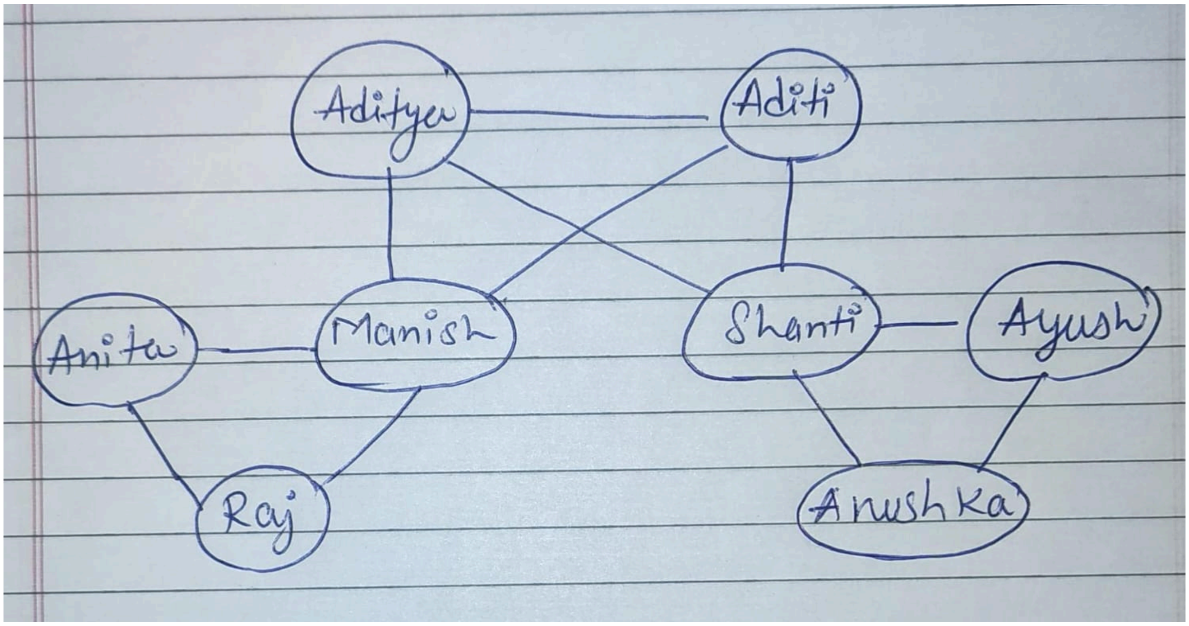
Activity 1: Example Family Tree/disease-symptom mapping/ City map with their distances between them:

Code:

```
male(Aditya).
male(Manish).
male(Raj).
male(Ayush).
female(Aditi).
female(Anita).
female(Shanti).
female(Ayush).

parent(Aditya, Manish).
parent(Aditi, Manish).

parent(Aditya, Shanti).
parent(Aditi, Shanti).

parent(Manish, Raj).
parent(Anita, Raj).

parent(Shanti, Ayush).
parent(Ayush, Ayush).

father(X, Y) :-
   parent(X, Y),
   male(X).

mother(X, Y) :-
   parent(X, Y),
   female(X).

sibling(X, Y) :-
   parent(Z, X),
   parent(Z, Y),
   X \= Y.
```

```
grandparent(X, Y) :-
    parent(X, Z),
    parent(Z, Y).

uncle(X, Y) :-
    sibling(X, Z),
    parent(Z, Y),
    male(X).

aunt(X, Y) :-
    sibling(X, Z),
    parent(Z, Y),
    female(X).

cousin(X, Y) :-
    parent(A, X),
    parent(B, Y),
    sibling(A, B),
    X \= Y.

grandchild(X, Y) :-
    grandparent(Y, X).

nephew_niece(X, Y) :-
    sibling(Y, Z),
    parent(Z, X).
```

Output:



Activity 2 : Implement a Goal based agent using PROLOG program
Problem statement:
Implement an AI-based investigation system that reconstructs a crime scenario by analyzing fragmented clues, suspect alibis, and event timelines, discard false leads, and arrive at the correct suspect who satisfies all constraints of the crime.

Story:
The Mystery of Stoke Moran
The Crime: Helen Stoner's sister, Julia, died under mysterious circumstances in a room that was locked from the inside. Two years later, Helen is sleeping in that same room and hears a low, clear whistle, the same sound her sister heard before dying.

The Fragmented Clues:

- The Ventilator: A small air vent connects the victim's room directly to Dr. Roylott's room.
- The Bell-rope: A rope hanging over the bed, which is not attached to any bell and ends just above the pillow.
- Exotic Animals: Dr. Roylott keeps a Cheetah and a Baboon that roam the grounds.
- The Safe: A metallic "clanging" sound was heard coming from Roylott's room at night.
- Milk on a Saucer: A small saucer of milk was found in Roylott's room, though he has no cat.

The Suspects & The Dead End

1. Dr. Roylott (The Primary Suspect)

- Motive: Financial inheritance; if the sisters marry, he loses control of their deceased mother’s fortune.
- Alibi: Claims to be asleep, but the ventilator and bell-rope provide a direct path into the locked room.
- Evidence: Knowledge of dangerous animals and the "clanging" safe where the "Speckled Band" (a swamp adder snake) is kept.

2. The Cheetah (The Dead End / False Lead)

- Context: The Cheetah is a dangerous, exotic animal wandering the grounds.
- The "Dead End": It explains why the sister was terrified and why strange sounds were heard on the lawn. However, a Cheetah cannot enter a locked room through a tiny ventilator or climb a bell-rope. It is a false lead designed to distract from the real murder weapon.

3. The Gypsies (The Secondary False Lead)

- Context: A band of gypsies is camping on the estate.
- The "Dead End": Julia's dying words were "The Speckled Band," which Helen initially thought referred to the spotted handkerchiefs worn by the gypsies. However, the gypsies have no way to enter the locked manor.

Code:
```
suspect(roylott).
suspect(sister).
suspect(gypsy).
suspect(cheetah).

motive(roylott, inheritance).
motive(sister, none).
motive(gypsy, none).
motive(cheetah, none).

access(roylott, ventilator).
access(sister, door).
access(gypsy, none).
access(cheetah, none).

state(door, locked).

status(X, dead_end) :-
    suspect(X),
    (motive(X, none) ; (access(X, door), state(door, locked))).

is_culprit(X) :-
```

```
    suspect(X),
    not(status(X, dead_end)),
    access(X, ventilator).

is_innocent(X) :- status(X, dead_end).
```

Output:

```
status(cheetah, Result).
Result = dead_end
Next | 10 | 100 | 1,000 | Stop

status(sister, Result)
Result = dead_end
Next | 10 | 100 | 1,000 | Stop

is_innocent(X).
X = sister
X = sister
X = gypsy
X = cheetah
false

is_culprit(X).
X = roylott
Next | 10 | 100 | 1,000 | Stop
```

Post Lab Subjective/Objective type Questions:

Question: Answer in brief

1. **Differentiate between a fact and a predicate with syntax.**
   A fact represents basic true information in PROLOG and does not contain variables. It is used to store data. For example, male(john). is a fact. A predicate defines a relationship or rule and usually contains variables to express logic. For example, father(X,Y) :- parent(X,Y), male(X). is a predicate that specifies a condition under which X is the father of Y.

2. **Differentiate between knowledgebase and Rule base approach.**
   A knowledgebase contains both facts and rules, representing complete domain knowledge and allowing reasoning through inference. A rule base contains only rules and focuses mainly on decision-making logic. The knowledgebase provides a broader representation of information,

whereas the rule base applies logical conditions to derive conclusions.

3. **Differentiate between database and knowledgebase.**
   A database stores raw data and requires explicit queries to retrieve information, but it does not support automatic reasoning. A knowledgebase stores structured knowledge including relationships and rules, allowing the system to infer new information. Databases focus on storage, while knowledgebases emphasize understanding and reasoning.

Question: Answer in detail.

Explain the role of PEAS and task environment in choosing the agent architecture. Justify your answer with an example.

PEAS stands for Performance measure, Environment, Actuators, and Sensors. It defines what an agent must achieve, where it operates, how it perceives the environment, and how it acts. Along with PEAS, the task environment characteristics such as observability, determinism, and dynamics help determine the appropriate agent architecture. For example, in a self-driving car, performance includes safety and speed, the environment consists of roads and traffic, sensors include cameras and radar, and actuators include brakes and steering. Since this environment is dynamic and unpredictable, a goal-based or utility-based agent is required rather than a simple reflex agent. Thus, PEAS and task environment analysis guide the selection of agent architecture, ensuring intelligent and flexible decision-making.

Conclusion:

This experiment successfully demonstrated the implementation of condition-action rule based and goal-based agents using PROLOG. The family tree program illustrated how facts and rules enable logical inference, while the goal-based investigation agent showed how constraints and goals can be used to reason toward a solution and eliminate false leads. The experiment helped in understanding the difference between reflex and goal-based agents, and highlighted PROLOG's effectiveness in representing knowledge and performing logical reasoning for AI applications.